Recommendations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected recommendation record.

Filter books, explain a preference tradeoff, inspect exposure and handle pending requirements.

Keep one habit: Keep the purpose, eligible candidates, scores and feedback context together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Similarity

Prediction Is Not Fact

Build a Human Classifier

W1 · Filter and score BookRec1

> Use the supplied records and explain each exclusion.

Your evidence record

- BookRec1 reference: require English and available. Score eligible books 2 for topic space and 1 for format comic; return all tied highest scores. J=space,prose,English,available; K=animals,comic,English,available; L=space,comic,Spanish,available; M=space,comic,English,unavailable. Unknown required facts mark a candidate pending verification. If none are verified eligible, report no verified eligible option and retain the pending reason.

- Apply BookRec1 to J, K, L and M from the lesson. State **eligibility** or exclusion reason for each, the scores of eligible books and the **recommendation**. Explain why L's two **preference** matches do not make it the winner.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Explain J's recommendation honestly

> Name both its matched preference and the one it lacks.

Your evidence record

- Write an explanation of why J was recommended that mentions one **preference** it does not match. A marketer adds guaranteed enjoyment. Explain why that phrase is unsupported.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Separate observed events from inferred feelings

> Retain what was shown and allow an unanswered feedback request.

Your evidence record

- J was shown and clicked. K was never shown. Separate the observed records from the labels enjoyed and disliked. Write a question that could gather a stated reason while allowing no answer.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Handle missing eligibility information

> Keep unknown availability pending instead of overriding a requirement.

Your evidence record

- Return to BookRec1. J and K become unavailable, L remains Spanish, and M's availability becomes unknown. A designer proposes recommending M anyway because it matches both **preference**s.

  Apply the requirements and propose an honest response.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Transfer the process to activities

> Keep one first attempt before hints or feedback.

Read and explain independently

A fictional learner needs an indoor activity lasting at most 20 minutes. **Preference**s score 2 for drawing and 1 for solo.

Candidates: A drawing, solo, outdoor, 10 minutes; B drawing, group, indoor, 20 minutes; C music, solo, indoor, 15 minutes; D drawing, group, indoor, 18 minutes. Filter, score and return all highest scorers.

Only B is then shown and selected; C and D are not shown. State what the feedback establishes and what remains unknown.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer the process to activities

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

If the sorter ranks bins, state the **objective** and show why the top suggestion is not a command.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Explain a **recommendation** honestly for **Recommendation**s and feedback. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Recommendation**s and feedback. Proposed explanation: “A high score can cancel a failed requirement”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A system shows only adventure books and records only adventure clicks. It concludes the reader dislikes poetry. Explain the missing **evidence** and propose a small test.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A recommender suggests adventure books after one adventure-book choice.

Explain its **evidence** and limitation.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A recommender suggests a space book because you liked two space books. Does the suggestion prove you will enjoy it? What feedback could help next time?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 20 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A system shows only adventure books and records only adventure clicks. It concludes the reader dislikes poetry. Explain the missing **evidence** and propose a small test.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A fictional learner needs an indoor activity lasting at most 20 minutes. **Preference**s score 2 for drawing and 1 for solo.

Candidates: A drawing, solo, outdoor, 10 minutes; B drawing, group, indoor, 20 minutes; C music, solo, indoor, 15 minutes; D drawing, group, indoor, 18 minutes. Filter, score and return all highest scorers.

Only B is then shown and selected; C and D are not shown. State what the feedback establishes and what remains unknown.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

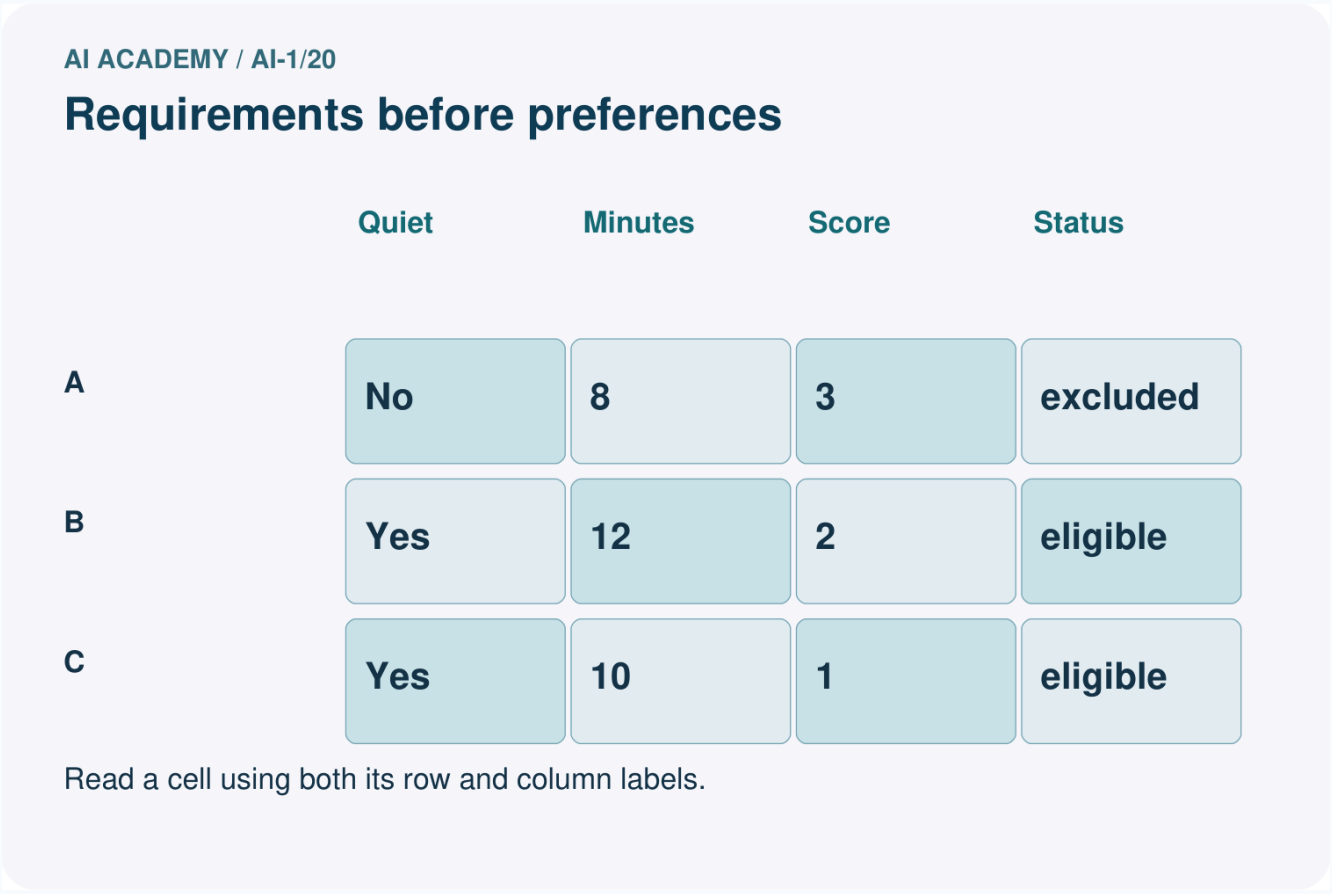
> Keep the explanation and the support you used.

Use this when helpful

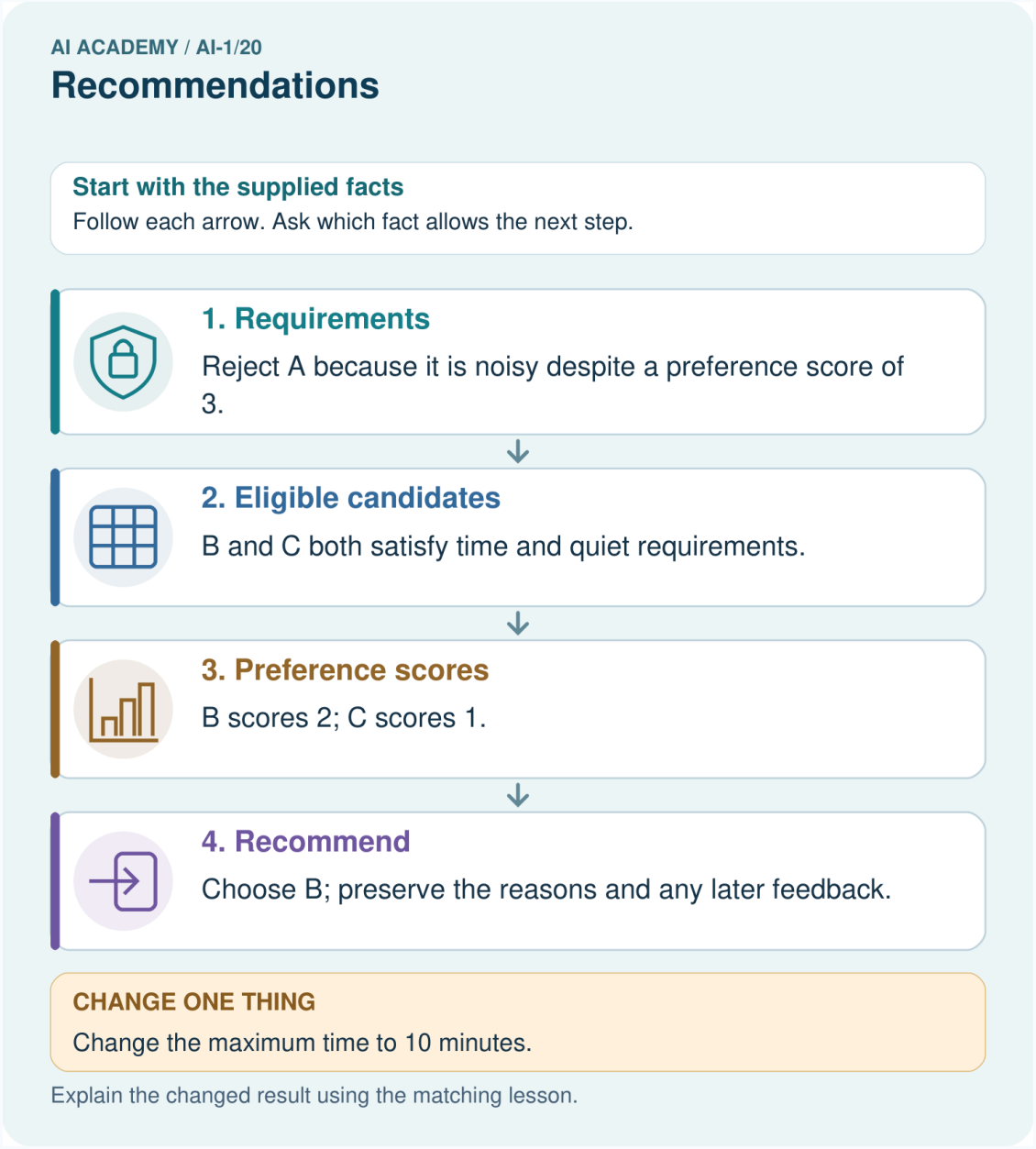
A fictional classroom break chooser recommends a suitable paper activity.

Must be quiet and take at most **12 minutes**. Score 2 for puzzles and 1 for solo.

A puzzle,solo,noisy,8 min; B puzzle,group,quiet,12 min; C drawing,solo,quiet,10 min.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why does the **recommendation** change without changing anyone's **preference**s?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional classroom break chooser recommends a suitable paper activity.
- Which labelled step matches this detail? Choose B; preserve the reasons and any later feedback.
- 1. Recommend
- 2. Requirements
- 3. Eligible candidates
- 4. **Preference** scores
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Recommend Choose B; preserve the reasons and any later feedback. A hard requirement determines whether an option can be chosen at all.

Preferences rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety.

Changed-case question

- Changed-case practice: Change the maximum time to 10 minutes. Which conclusion is supported by the supplied facts?
- 1. Only C remains eligible and is recommended.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Only C remains eligible and is recommended. B takes **12 minutes**; A is still noisy.

Repair and save your work

Review Similarity (ai-1/19). Goal: Calculate similarity using explicitly selected features and a stated scoring rule. Short practice: Point to the matches supporting C's new score and explain why there is no tie..

Return to this week and explain what you changed.

Review **Prediction** Is Not Fact (ai-1/17). Goal: Separate a recorded **prediction** from the **observation** used to check it. Short practice: Which field changes the **observation** label, and which field changes the model **prediction**?.

Return to this week and explain what you changed.

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short..
Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

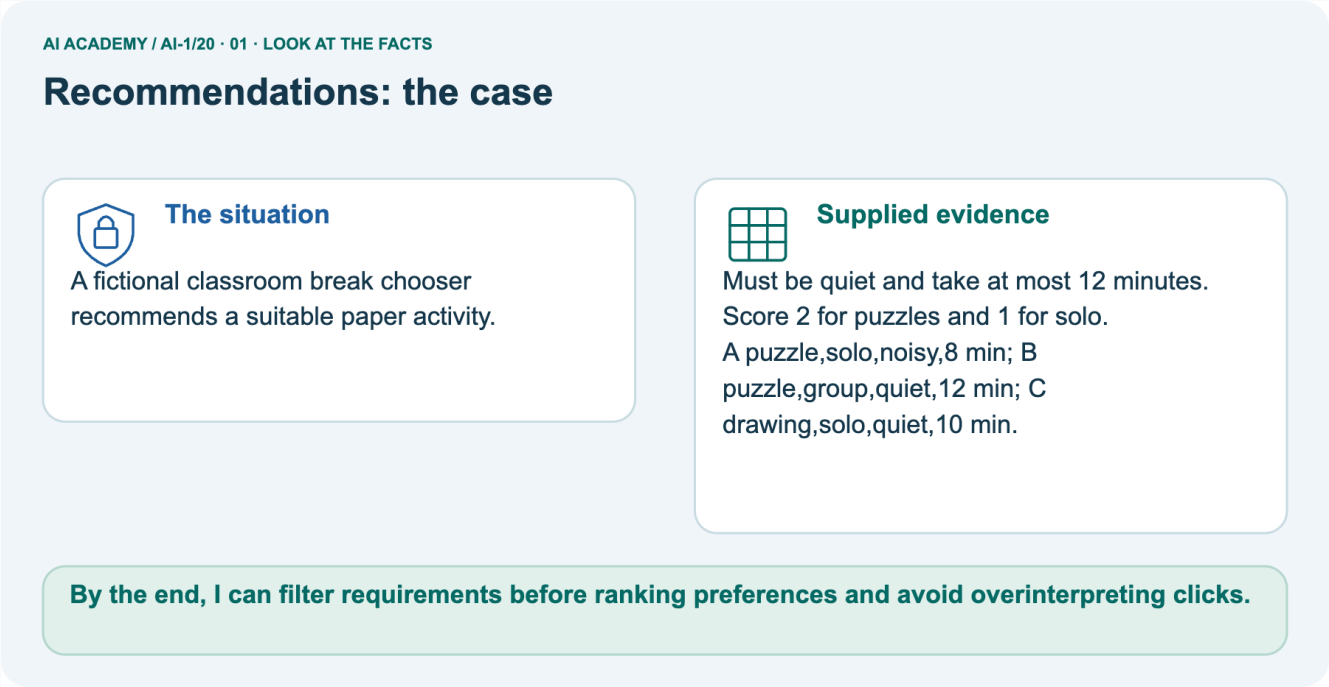
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Recommendations: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

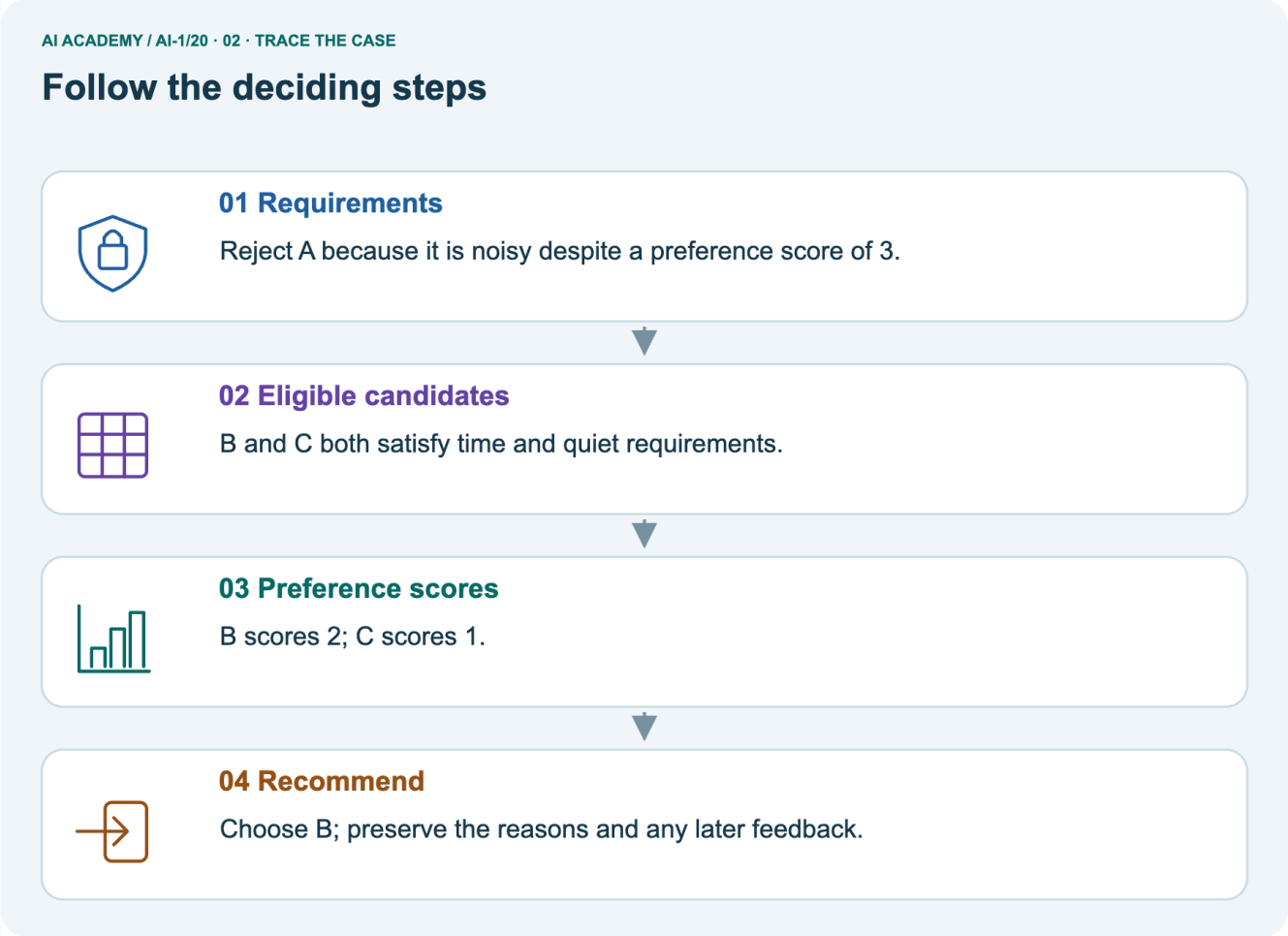
A fictional classroom break chooser recommends a suitable paper activity. Supplied facts: Must be quiet and take at most **12 minutes**. Score 2 for puzzles and 1 for solo.

A puzzle,solo,noisy,8 min; B puzzle,group,quiet,12 min; C drawing,solo,quiet,10 min.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



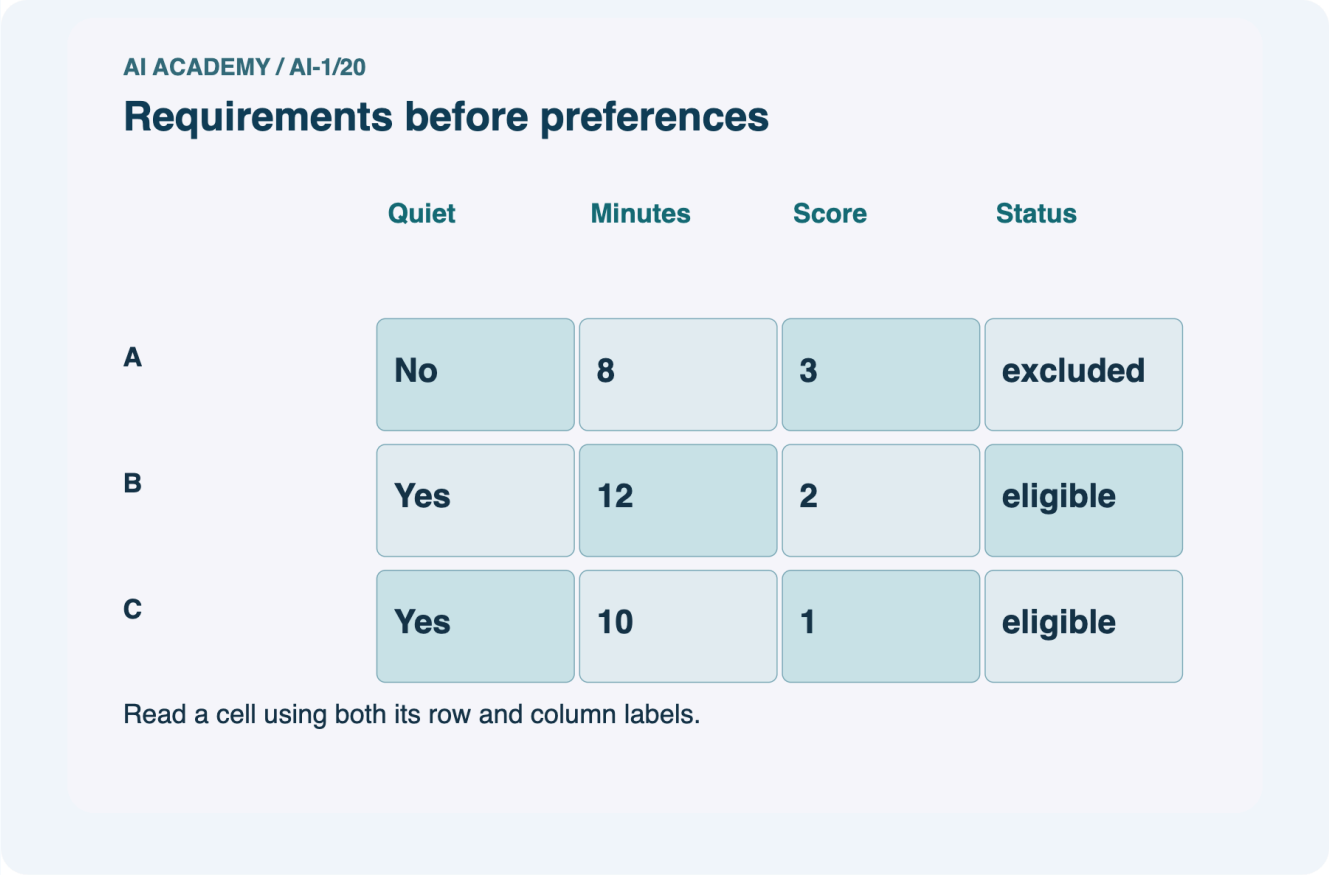
Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements: Reject A because it is noisy despite a **preference** score of 3. → Eligible candidates: B and C both satisfy time and quiet requirements. → **Preference** scores: B scores 2; C scores 1. → Recommend: Choose B; preserve the reasons and any later feedback.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Requirements before preferences



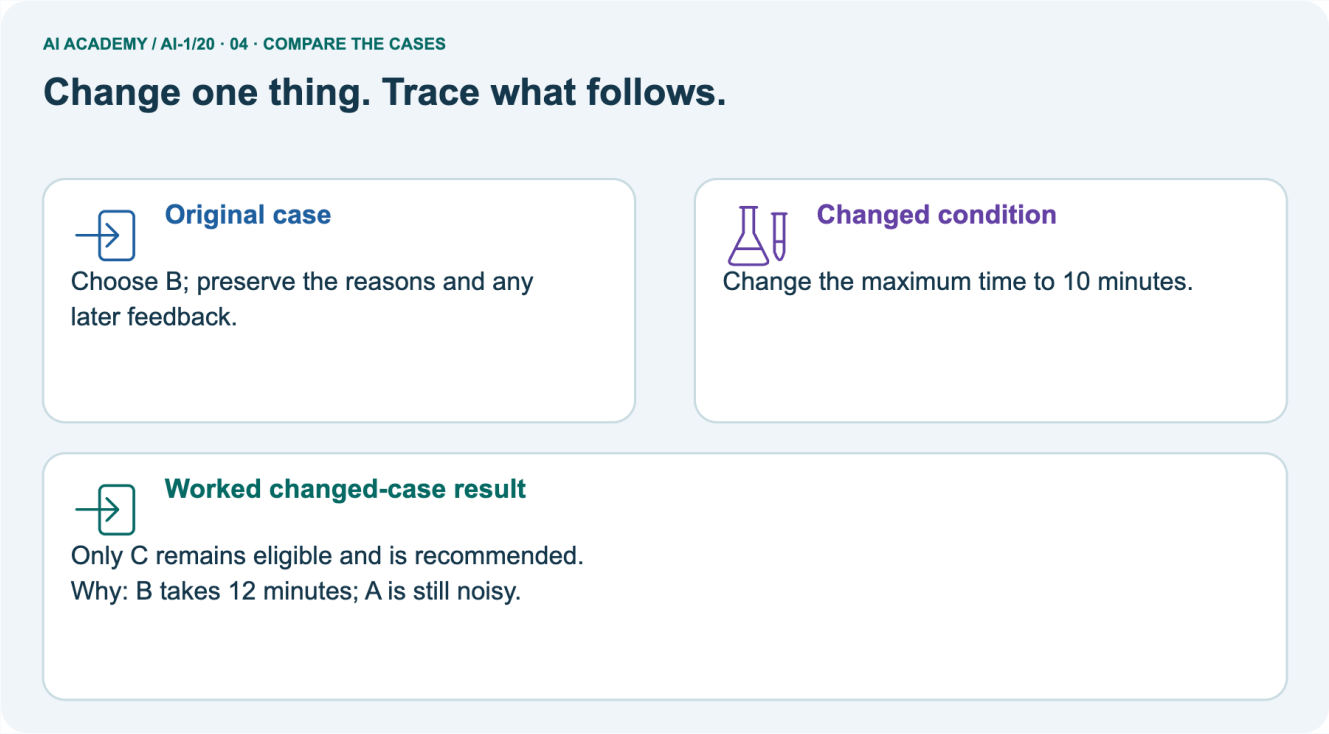
Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements before **preference**s. A: Quiet = No, Minutes = 8, Score = 3, Status = excluded. B: Quiet = Yes, Minutes = 12, Score = 2, Status = eligible. C: Quiet = Yes, Minutes = 10, Score = 1, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

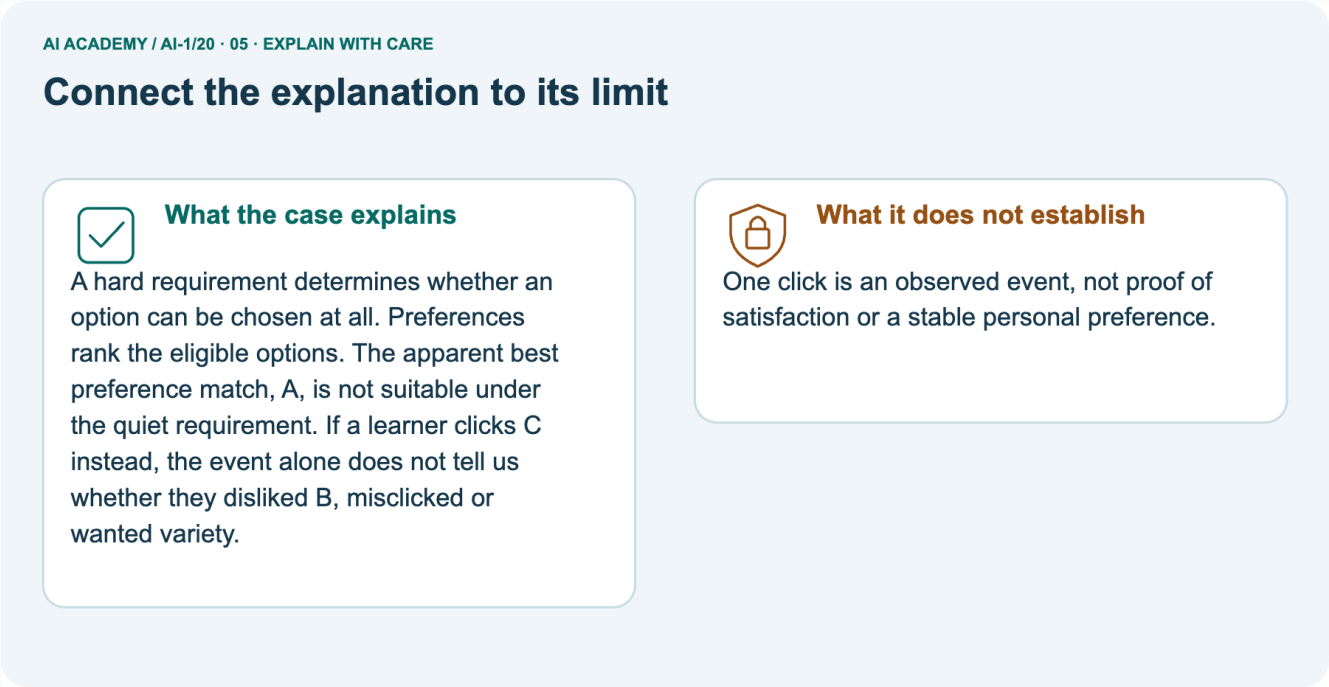
Original case: Choose B; preserve the reasons and any later feedback. Changed condition: Change the maximum time to 10 minutes. Expected result: Only C remains eligible and is recommended.

Reason: B takes **12 minutes**; A is still noisy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

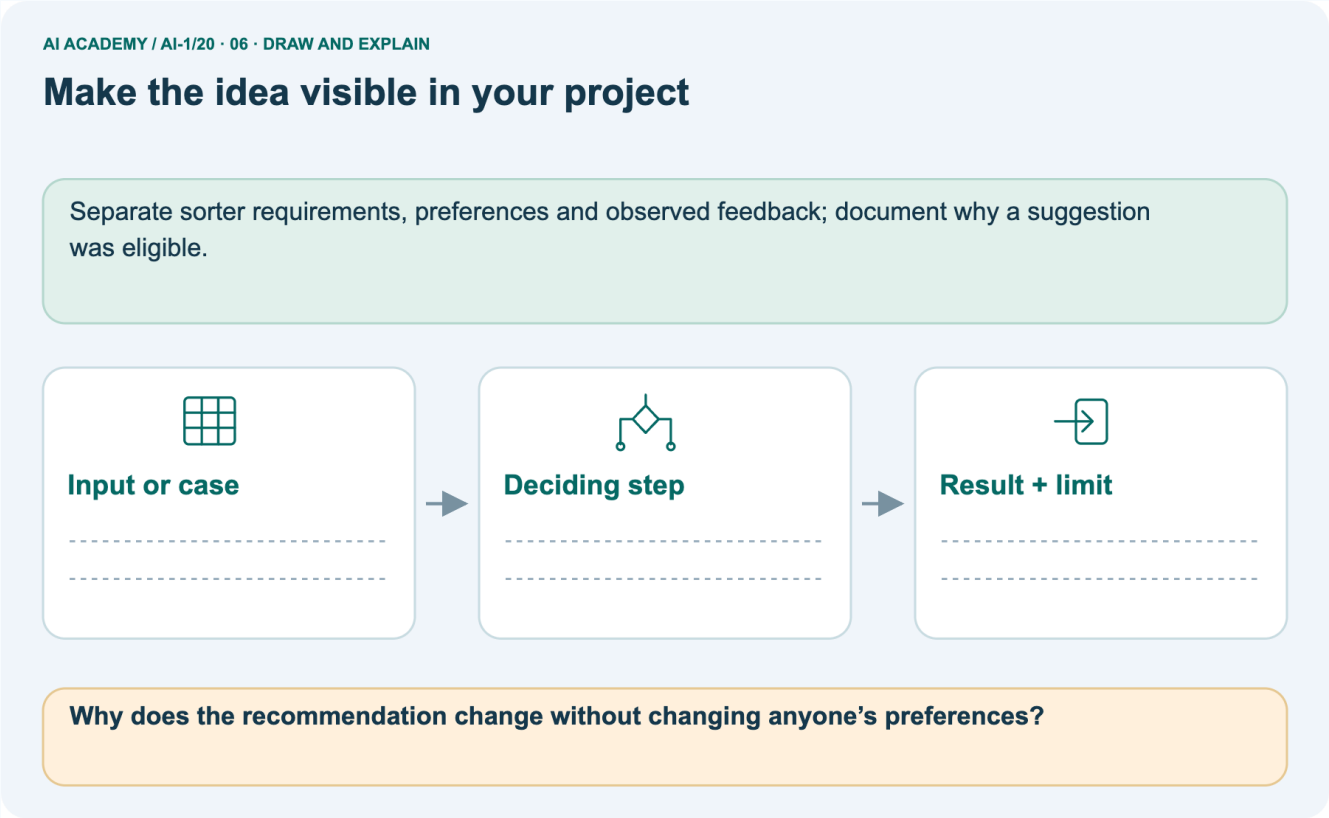
A hard requirement determines whether an option can be chosen at all. **Preference**s rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety. Limit: One click is an observed event, not proof of satisfaction or a stable personal **preference**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible. Draw your **input**, deciding step, result and limit.

Supported question: Why does the **recommendation** change without changing anyone's **preference**s?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Recommendations: the case

02. Follow the deciding steps

03. Requirements before preferences

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.